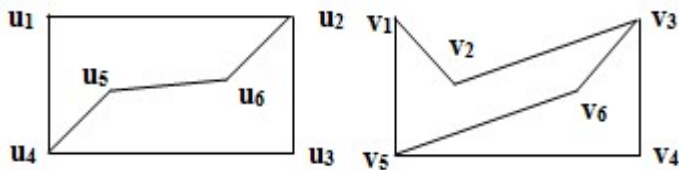
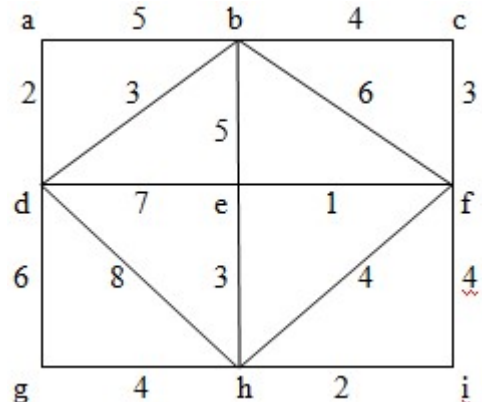


Q.No.	Questions	M	CO	KL
UNIT-I				
1.a)	Establish that $\{((p \vee q) \rightarrow r) \wedge (\neg p)\} \rightarrow (q \rightarrow r)$ is a tautology	7	CO1	K2
b)	Establish that the following argument is valid by using the rules of inference If Clifton does not live in France, then he does not speak French. Clifton does not drive a Datsun If Clifton lives in France, then he rides a bicycle Either Clifton speaks French, or he drives a Datsun Hence, Clifton rides a bicycle	7	CO1	K3
(OR)				
2.a)	Verify that the following argument is valid by using the rules of inference, quantifiers. Babies are illogical. Nobody is despised who can manage a crocodile. Illogical people are despised. Hence, babies cannot manage crocodiles.	7	CO1	K3
b)	Determine the PDNF and PCNF of $p \rightarrow [(p \rightarrow q) \wedge \sim(\sim q \vee \sim p)]$	7	CO1	K3
UNIT-II				
3.a)	Determine the number of ways of arranging 6 boys and 6 girls in a row. Also determine the number of arrangements in which i) All girls will be together. ii) No two girls will be together. iii) Boys and girls come alternatively.	7	CO2	K2
b)	i) Determine the term independent of x in the expansion of $(x^2 + \frac{1}{x})^{12}$ ii) Determine the coefficient of $x^5 y^{10} z^5 w^5$ in the expansion $(x + 7y + 3z + w)^{25}$	7	CO2	K2
(OR)				
4.a)	A cricket team of 11 is to be selected out of 14 players of whom 5 are bowlers. Determine the number of ways in which this can be done so as to include at least 3 bowlers.	7	CO2	K2
b)	Determine the number of integers between 1 and 250 which are divisible by any of the integers 2, 3, 5 or 7.	7	CO2	K3
UNIT-III				
5.a)	Let R denote a relation on the set of ordered pairs of positive integers such that $(x, y)R(u, v)$ if and only if $xv = yu$. Then establish that 'R' is an equivalence relation.	7	CO3	K2

b)	Define Poset and draw Hasse diagrams for the Posets (i) $[D_{12};/]$ (ii) $[\{2,3,5,30,60,120,180,360\};/]$ and (iii) $[P(A),\subseteq]$, where $A = \{1,2,3\}$	7	CO3	K2
(OR)				
6.a)	Establish that a Lattice 'L' is distributive iff $\forall x, y, z \in L \quad (x * y) \oplus (y * z) \oplus (z * x) \equiv (x \oplus y) * (y \oplus z) * (z \oplus x)$	7	CO4	K3
b)	Consider the Boolean polynomial $p(x, y, z) = x * (y \oplus z')$. If $B = \{0,1\}$, compute the truth table of the function $f: B_3 \rightarrow B$ defined by p . Also draw logical diagram.	7	CO4	K3
UNIT-IV				
7.a)	Determine the number of integral solutions for the equation $x_1 + x_2 + x_3 + x_4 + x_5 = 20$ where $x_1 \geq 3, x_2 \geq 2, x_3 \geq 4, x_4 \geq 6$ and $x_5 \geq 0$.	7	CO5	K3
b)	Solve the recurrence relation $a_n - 7a_{n-1} + 10a_{n-2} = 4^n$ for $n \geq 2$	7	CO5	K3
(OR)				
8.a)	i) Calculate $B(X) = \sum_{r=0}^{\infty} b_r X^r = \frac{1}{X^2 - 6X + 8}$ ii) Find the coefficient of X^{20} in $(X^3 + X^4 + X^5 + \dots)^5$	7	CO5	K2
b)	Solve the recurrence relation $a_n - 5a_{n-1} + 6a_{n-2} = 0, n \geq 2$ by using generating functions.	7	CO5	K3
UNIT-V				
9.a)	Determine the Isomorphism between the following graphs. 	7	CO6	K3
b)	State and Prove Euler's formula for planar graphs.	7	CO6	K2
(OR)				
10.a)	Establish that a tree with 'n' elements has exactly 'n-1' edges.	7	CO6	K2
b)	Explain Kruskal's algorithm and using it find the minimal spanning tree for the following weighted graph 	7	CO6	K3

Course Code: B20IT2101					
SAGI RAMAKRISHNAM RAJU ENGINEERING COLLEGE (A)				R20	
II B. Tech I Semester Regular Examinations					
DATA STRUCTURES					
(Common to AIDS & IT)					
Time: 3 Hrs.				Max. Marks:70	
Answer any one Question from Each Unit					
All questions carry equal Marks					
			CO	KL	M
UNIT-I					
1	a)	Define data structure. Discuss different types of data structure their implementations applications.	CO1	K2	7
	b)	Explain the recursive merge sort algorithm to sort the following elements: 12, 25, 5, 9, 1, 84, 63, 7, 15, 4, 3.	CO1	K2	7
OR					
2	a)	Write Recursive Binary Search algorithm. Search element 12 in the given list using Recursive Binary Search algorithm 2, 5, 12, 15, 23, 28, 36, 39, 45	CO1	K2	7
	b)	Rearrange following numbers using quick sort: 10, 6, 3, 7, 17, 26, 56, 32, 72	CO1	K3	7
UNIT-II					
3	a)	Explain the procedure to evaluate postfix expression. Evaluate the following postfix expression 7 3 4 + - 2 4 5 / + * 6 / 7 +	CO2	K2	7
	b)	Discuss the algorithms for push and pop operations on a stack	CO2	K2	7
OR					
4	a)	Give the structure of Queue ADT. Explain the operations in it.	CO2	K2	7
	b)	Convert the given infix expression $A+B^{\wedge}C+(D^{\wedge}E/F)^{\wedge}G$ into its postfix expression, and evaluate the same using stack. Here A=3, B=5, C=2, D=7, E=4, F=1, G=8.	CO2	K3	7
UNIT-III					
5	a)	Write an algorithm for representing the polynomial $6x^0 + 4x^3 - 2x + 10$ using linked lists.	CO2	K3	7
	b)	Write an algorithm to insert new node at the beginning and at the end of a singly linked list.	CO2	K3	7
OR					
6	a)	Write an algorithm to push and pop an element from linked stack	CO2	K3	7
	b)	Write an algorithm to delete an element anywhere from doubly linked list.	CO2	K3	7
UNIT-IV					
7	a)	Explain Heap sort algorithm. Create Heap for the following elements and then sort them. (13, 102, 405, 136, 15, 105, 390, 432, 28, 444)	CO3	K3	7
	b)	Develop a binary search tree resulting after inserting the following integer keys 49, 27, 12, 11, 33, 77, 26, 56, 23, 6. (i) Check whether the tree is almost complete or not? (ii) Determine the height of the tree (iii) Write post order and pre-order traversals	CO3	K3	7
OR					
8	a)	How to represent binary tree using arrays and linked list?	CO3	K2	7
	b)	Construct an AVL tree by inserting the following elements successively C O M P U T E R	CO3	K2	7

UNIT-V					
9	a)	Explain Warshall's algorithm to find transitive closure of a graph with a suitable example.	CO3	K2	7
	b)	What is minimum cost spanning Tree? Explain the process of finding the minimum spanning tree with suitable example.	CO3	K2	7
OR					
10	a)	Explain Depth First Search algorithms in detail.	CO3	K2	7
	b)	Explain Dijkstra's algorithm with suitable example.	CO3	K2	7

Course Code:B20AD2101					
SAGI RAMAKRISHNAM RAJU ENGINEERING COLLEGE (A)				R20	
II B. Tech I Semester Regular Examinations					
INTRODUCTION TO ARTIFICIAL INTELLIGENCE					
(For AIDS)					
Time: 3 Hrs.				Max. Marks:70	
Answer any one Question from Each Unit					
All questions carry equal Marks					
			CO	KL	M
UNIT-I					
1	a)	List out various applications of AI.	CO1	K2	7
	b)	What are the advantages & disadvantages of AI?	CO1	K3	7
OR					
2	a)	Elaborate the implementation of Tic-Tac-Toe game with 3 approaches.	CO1	K3	7
	b)	Categorize intelligent systems based on their working principle.	CO1	K3	7
UNIT-II					
3	a)	Write the productions involved in solving a Water-Jug Problem.	CO2	K3	7
	b)	Elaborate the working of A* Algorithm with an example.	CO2	K3	7
OR					
4	a)	Explain the procedure to implement Hill Climbing.	CO2	K3	7
	b)	Discuss the implementation of all the exhaustive searches with examples.	CO2	K3	7
UNIT-III					
5	a)	Trace the Resolution Algorithm by taking an example.	CO3	K3	7
	b)	Discuss the procedure of converting WFF to the clause form.	CO3	K3	7
OR					
6	a)	Explain Propositional Calculus (PC).	CO3	K2	7
	b)	Explain Propositional Logic (PL).	CO3	K2	7
UNIT-IV					
7	a)	Discuss the procedure to represent knowledge using Semantic Network.	CO4	K3	7
	b)	How do you represent visiting a restaurant in the form of a Script? Explain	CO4	K3	7
OR					
8	a)	Write the significance of using CYC in capturing human commonsense database.	CO4	K3	7
	b)	Write about Conceptual Dependency theory. How it will be used for Knowledge Representation?	CO4	K3	7
UNIT-V					
9	a)	Differentiate Expert Systems versus Traditional Systems.	CO5	K3	7
	b)	Write the significance of Bayes’s theorem in AI	CO5	K3	7
OR					
10	a)	Explain Dempster-Shafer Theory	CO5	K3	7
	b)	Explain the significance of various Fuzzy Set Operations.	CO5	K3	7

Course Code: B20IT2103					
SAGI RAMAKRISHNAM RAJU ENGINEERING COLLEGE (A)					R20
II B. Tech I Semester Regular Examinations					
PYTHON PROGRAMMING					
(Common to AIDS & IT)					
Time: 3 Hrs.					Max. Marks:70
Answer any one Question from Each Unit					
All questions carry equal Marks					
			CO	KL	M
UNIT-I					
1	a)	Explain about input validation loops and nested loops with examples	CO1	K3	7
	b)	Write a Python program to calculate the amount payable if money has been lent on simple interest. Principal or money lent = P, Rate of interest = R% per annum and Time = T years. Then Simple Interest (SI) = (P x R x T)/ 100. Amount payable = Principal + SI. P, R and T are given as input to the program.	CO1	K4	7
OR					
2	a)	Explain about explicit conversion with examples.	CO1	K2	7
	b)	Explain about precedence of all operators in Python.	CO1	K2	7
UNIT-II					
3	a)	Define Python string padding functions? Explain with examples	CO2	K2	7
	b)	Illustrate if, if-else, if-elif-else Statements with examples.	CO2	K3	7
OR					
4	a)	Explain about data encryption in Python.	CO2	K2	7
	b)	Presume that a ladder is put upright against a wall. Let variables length and angle store the length of the ladder and the angle that it forms with the ground as it leans against the wall. Write a Python program to compute the height reached by the ladder on the wall for the following values of length and angle: i) 16 feet and 75 degrees ii) 20 feet and 0 degrees iii) 24 feet and 45 degrees iv) 24 feet and 80 degrees	CO2	K3	7
UNIT-III					
5	a)	Write a Python program to create three dictionaries, then create one dictionary that will contain the other three dictionaries.	CO3	K3	7
	b)	Describe Python list/Array methods? Explain.	CO3	K2	7
OR					
6	a)	Discuss about importing module from a package.	CO3	K2	7
	b)	Explain about anonymous or Lambda function with merits and demerits	CO3	K2	7
UNIT-IV					
7	a)	Explain about structuring classes with inheritance and polymorphism.	CO4	K2	7
	b)	Illustrate manipulating file pointer using seek with suitable example.	CO4	K3	7
OR					
8	a)	Demonstrate the case study of an ATM using classes.	CO4	K3	7

	b)	Explain about reading numbers from a file using Python program.	CO4	K2	7
UNIT-V					
9	a)	Defining clean-up actions .	CO5	K2	7
	b)	Illustrate Entry fields for the input and output of text with example.	CO5	K3	7
OR					
10	a)	Describe user Defined exception with example.	CO5	K2	7
	b)	Define Scrolling list boxes With example .	CO5	K2	7

				Code: B20AD2102	
SAGI RAMAKRISHNAM RAJU ENGINEERING COLLEGE (A)				R20	
II B. Tech I Semester Regular Examinations					
COMPUTER ORGANIZATION					
(For AIDS)					
Time: 3 Hrs.			Max. Marks:70		
Answer any one Question from Each Unit					
All questions carry equal Marks					
			M	CO	KL
1.	(a).	Explain about Complements and uses of complements	7M	CO1	K2
	(b).	Describe Data Representation in basic computers	7M	CO1	K2
(OR)					
2.	(a).	Describe Multiplication Algorithm with example	7M	CO1	K2
	(b).	Explain Error Detection Codes	7M	CO1	K2
UNIT-II					
3.	(a).	Describe Instruction cycle in computer system	7M	CO1	K2
	(b).	Explain about Bus and Memory Transfer	7M	CO1	K2
(OR)					
4.	(a).	Design Binary Adder and Subtractor and explain	7M	CO1	K2
	(b).	Explain about Computer instructions	7M	CO1	K2
UNIT-III					
5.	(a).	Explain about General register organization with seven registers	7M	CO2	K2
	(b).	Expand the given statement in Three,Two,One ,Zero Addresses A=(B+C)*(D+E)	7M	CO2	K3
(OR)					
6.	(a).	Describe the Addressing Modes i) Direct ii) Relative iii) Auto Increment	7M	CO2	K2
	(b).	Explain about micro programmed control.	7M	CO2	K1
UNIT-IV					
7.	(a).	Discuss about Virtual memory	7M	CO3	K2
	(b).	Explain about Asynchronous Communication interface with neat diagram	7M	CO3	K2
(OR)					
8.	(a).	Explain about Asynchronous Data transfer	7M	CO3	K2
	(b).	Explain Memory Mapping Techniques of Cache Memory	7M	CO3	K2
UNIT-V					
9.	(a).	Describe the Characteristics of Multiprocessors	7M	CO4	K2
	(b).	Explain about Array Processor.	7M	CO4	K2
(OR)					
10.	(a).	Describe Interconnection Structures with neat diagram	7M	CO4	K2
	(b).	Describe about Pipeline and vector processing	7M	CO4	K2

Time: 3 Hrs.

Max. Marks:70

Answer any one Question from Each Unit

All questions carry equal Marks

			CO	KL	M
UNIT-I					
1	a)	Define algorithm. Explain asymptotic notations Big O, Omega, and Theta.	CO1	K3	7
	b)	Write an algorithm for matrix multiplication and find its time complexity.	CO1	K2	7
OR					
2	a)	What is an articulation point? Explain the procedure to determine bi-connected components in the graph with example.	CO1	K5	7
	b)	Write an algorithm for BFS. Explain with example.	CO1	K4	7
UNIT-II					
3	a)	Explain divide-and-conquer technique. Write a recursive algorithm for finding the maximum and minimum element from the list.	CO2	K4	7
	b)	Illustrate the tracing of Quick Sort algorithm for the following set of numbers 25, 10, 72, 18, 40, 11, 64, 58, 32, 9	CO2	K4	7
OR					
4	a)	Find the optimal solution of the knapsack instance $n = 7$, $M = 15$, $(p_1, p_2, \dots, p_7) = (10, 5, 15, 7, 6, 18, 3)$ and $(w_1, w_2, \dots, w_7) = (2, 3, 5, 7, 1, 4, 1)$.	CO2	K2	7
	b)	What is job sequencing with deadlines problem? Let $n=5$, profit (10, 3, 33, 11, 40) and deadlines (3, 1, 1, 2, 2) respectively. Find the optimal solution using greedy method.	CO2	K3	7
UNIT-III					
5	a)	For the given cost matrix, obtain an optimal cost tour using dynamic programming 0101520 50910 613012 8890	CO3	K3	7
	b)	What is graph coloring? Write an algorithm for it and explain with an example.	CO3	K2	7
OR					
6	a)	Using dynamic programming, solve the following knapsack instance $n=4$, $m=5$, $(W_1, W_2, W_3, W_4)=(2, 1, 3, 2)$, $(P_1, P_2, P_3, P_4)=(12, 10, 20, 15)$.	CO3	K2	7
	b)	Explain Multistage graphs with example. Write multistage graph algorithm to forward approach.	CO3	K4	7
UNIT-IV					
7	a)	What is backtracking? Apply backtracking to solve the instance of the sum of subset problem $n=6$, $d=30$, $S=\{5, 10, 12, 13, 15, 18\}$	CO3	K4	7
	b)	Explain backtracking concept. Illustrate N queens problem using backtracking to solve 4-Queens problem.	CO3	K4	7
OR					

8	a)	State travelling salesperson problem. Apply Branch and Bound algorithm to solve the TSP instantiated by the following cost matrix. ∞ 20 30 10 11 15 ∞ 16 4 2 3 5 ∞ 2 4 19 6 18 ∞ 3 16 4 7 16 ∞	CO3	K4	7
	b)	Explain FIFO Branch and Bound	CO3	K2	7
UNIT-V					
9	a)	Explain the classes of NP-Hard and NP-complete.	CO4	K4	7
	b)	Apply the Rabin-Karp algorithm to search for the pattern AABA in the text A A B A A C A A D A A B A A B A	CO4	K2	7
OR					
10	a)	Explain the classes P and NP	CO4	K3	7
	b)	Apply the Knuth-Morris-Pratt matching to search for the pattern BAOBAB in the text BESS_KNEW_ABOUT_BAOBABS	CO4	K3	7

Course Code:B20BS2201																													
SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (A)						R 20																							
II B. Tech II Semester Regular Examinations																													
PROBABILITY AND STATISTICS																													
(Common to AIDS & CSE)																													
Time: 3 Hrs.			Max. Marks:70																										
Answer any one Question from Each Unit																													
All questions carry equal Marks																													
Q.No	Questions					CO	KL	M																					
1.a)	What are the different methods in collecting Primary and Secondary data?					CO1	K2	7																					
b)	Lives of two models of Refrigerators in a research survey are given below. <table border="1"><tr><td>Lives in Year</td><td>0 -2</td><td>2 - 4</td><td>4 - 6</td><td>6 - 8</td><td>8 -12</td><td>12 -14</td></tr><tr><td>Model - A</td><td>2</td><td>16</td><td>13</td><td>7</td><td>5</td><td>4</td></tr><tr><td>Model - B</td><td>2</td><td>7</td><td>12</td><td>19</td><td>9</td><td>1</td></tr></table> Determine the average life of each model and which model has greater uniformity.					Lives in Year	0 -2	2 - 4	4 - 6	6 - 8	8 -12	12 -14	Model - A	2	16	13	7	5	4	Model - B	2	7	12	19	9	1	CO1	K3	7
Lives in Year	0 -2	2 - 4	4 - 6	6 - 8	8 -12	12 -14																							
Model - A	2	16	13	7	5	4																							
Model - B	2	7	12	19	9	1																							
(OR)																													
2.a)	Define Random variable and explain about types of random variables with examples. Also discuss discrete and continuous probability functions of random variables.					CO1	K2	7																					
b)	A continuous random variable X has a pdf $f(x) = 3x^2$, $0 < x < 1$. Determine a and b such that (i) $P(X \leq a) = P(X > a)$ and (ii) $P(X > b) = 0.05$.					CO1	K3	7																					
3.a)	A two-dimensional r.v. (X, Y) has a bivariate distribution given by $P(X, Y) = \frac{x^2 + y}{32}$, for $x = 0, 1, 2, 3$ and $y = 0, 1$. Calculate marginal distributions of X and Y.					CO2	K3	7																					
b)	Let X be a random variable with the following distribution <table border="1"><tr><td>X</td><td>6</td><td>9</td><td>12</td></tr><tr><td>P(X=x)</td><td>1/6</td><td>1/2</td><td>1/3</td></tr></table> Determine $E(x)$, $E(x^2)$ and using the laws of expectation calculate $E(2X+1)^2$					X	6	9	12	P(X=x)	1/6	1/2	1/3	CO2	K3	7													
X	6	9	12																										
P(X=x)	1/6	1/2	1/3																										
(OR)																													
4. a)	A random variable X has the probability density function given by $f(x) = \begin{cases} x & \text{if } 0 \leq x < 1 \\ 2 - x & \text{if } 1 \leq x < 2 \\ 0 & \text{elsewhere} \end{cases}$ Determine $E(X)$ and $V(X)$.					C02	K3	7																					
b)	Define moment generating function and write its properties					C02	K2	7																					

5.a)	Determine a 2 nd degree regressed polynomial $y = a + b x + c x^2$ for the following data										CO3	K3	7	
	x	0	1	2	3	4								
	y	1	1.5	2.6	4.2	6.8								
b)	Determine the regression lines of x on y for the following data										CO3	K3	7	
	X	1	2	3	4	5	6							
	Y	15	17	14	18	16	15							
(OR)														
6.a)	Describe the procedure for fitting an exponential curve of the form $y = ax^b$										CO4	K3	7	
b)	Calculate the rank correlation between male and female ranks in a game										CO4	K2	7	
	Male ranks		3	5	1	4	2	7	6	8				
	Female ranks		4	2	3	1	5	6	8	7				
7.a)	Ten coins are thrown simultaneously. Determine the probability of getting at least i) seven heads ii) six heads										C05	K3	7	
b)	Fit a Poisson distribution for the following data										C05	K2	7	
	x	0	1	2	3	4	5	6	7	8				
	f(x)	56	156	132	92	37	22	4	0	1				
(OR)														
8. a)	In a normal distribution 31% of the items are under 45 and 8% are over 64. Determine mean and standard deviation										C05	K3	7	
b)	Derive mean and variance of Uniform distribution										C05	K3	7	
9.a)	Explain the following concepts (i) Null and Alternative Hypothesis (ii) Type I and Type II errors (iii) Critical region and level of significance.										C06	K1	7	
b)	Before an increase in excise duty on tea, 800 persons out of a sample of 1,000 persons were found to be tea drinkers. After an increase in duty, 800 people were tea drinkers in a sample of 1,200 people. Using standard error proportion, state whether there is a significant decrease in the consumption of tea after the increase in excise duty.										C06	K2	7	
(OR)														
10.a)	The height of 10 males of a given locality are found to be 70, 67, 62, 68, 61, 68, 70, 64, 64, 66 inches, Is it reasonable to believe that the average height is greater than 64 inches? Test at 5% significance level assuming that for 9 degrees of freedom $P(t>1.83) = 0.05$.										C06	K2	7	
b)	The following figures show the distribution of digits in numbers chosen at random from a telephone directory.										C06	K3	7	
	Digits	0	1	2	3	4	5	6	7	8				9
	frequency	1026	1107	997	966	1075	933	1107	972	964				853
Test whether the digits may be taken to occur equally frequently in the directory.														

Course Code: B20AD2202																																	
SAGI RAMAKRISHNAM RAJU ENGINEERING COLLEGE (A)				R20																													
II B. Tech II Semester Regular Examinations																																	
OPERATING SYSTEMS																																	
(For AIDS)																																	
Time: 3 Hrs.			Max. Marks:70																														
Answer any one Question from Each Unit																																	
All questions carry equal Marks																																	
			CO	KL	M																												
UNIT – I																																	
1.	a).	List and Describe the major services provided by Operating System	1	K2	7																												
	b).	Why Micro Kernel Structure is better than Layered Structure? Explain.	1	K2	7																												
OR																																	
2.	a).	Discuss the purpose of Interrupts? How the Interrupts are handled by Operating System?	1	K2	7																												
	b).	What is Debugging? Explain OS debugging?	1	K2	7																												
UNIT – II																																	
3.	a).	The following table represents details of four processes <table><tr><td>Process</td><td>Arrival Time</td><td>Burst Time</td></tr><tr><td>P1</td><td>0</td><td>6</td></tr><tr><td>P2</td><td>4</td><td>8</td></tr><tr><td>P3</td><td>6</td><td>7</td></tr><tr><td>P4</td><td>2</td><td>3</td></tr></table> Calculate average waiting time and Average Turnaround time using the following scheduling algorithms i) Non-Preemptive Shortest Job First Scheduling algorithm ii) Preemptive Shortest Job First Scheduling algorithm	Process	Arrival Time	Burst Time	P1	0	6	P2	4	8	P3	6	7	P4	2	3	2	K3	8													
Process	Arrival Time	Burst Time																															
P1	0	6																															
P2	4	8																															
P3	6	7																															
P4	2	3																															
	b).	Explain any two Multi Processor scheduling algorithms? List out its advantages and limitations	2	K2	6																												
OR																																	
4.	a).	The following table represents details of four processes <table><tr><td>Process</td><td>Arrival Times</td><td>Burst Time</td><td>Priority</td></tr><tr><td>P1</td><td>8</td><td>9</td><td>4</td></tr><tr><td>P2</td><td>6</td><td>5</td><td>5</td></tr><tr><td>P3</td><td>4</td><td>6</td><td>1</td></tr><tr><td>P4</td><td>4</td><td>5</td><td>6</td></tr><tr><td>P5</td><td>0</td><td>6</td><td>3</td></tr><tr><td>P6</td><td>2</td><td>8</td><td>2</td></tr></table> Calculate average waiting time and Average Turnaround time using the following scheduling algorithms i) Non-Preemptive Priority scheduling algorithm ii) Preemptive Priority scheduling algorithm	Process	Arrival Times	Burst Time	Priority	P1	8	9	4	P2	6	5	5	P3	4	6	1	P4	4	5	6	P5	0	6	3	P6	2	8	2	2	K3	8
Process	Arrival Times	Burst Time	Priority																														
P1	8	9	4																														
P2	6	5	5																														
P3	4	6	1																														
P4	4	5	6																														
P5	0	6	3																														
P6	2	8	2																														
	b).	Identify the different issues in thread scheduled?	2	K2	6																												
UNIT – III																																	
5.	a).	Explain the following terms with suitable examples i) Critical Section ii) Semaphore	3	K2	6																												
	b).	Design a solution for handling deadlock when it is occurred?	3	K3	8																												

OR					
6.	a).	Write a semaphore solution for the Readers and Writers problem	3	K3	7
	b).	Design a deadlock avoidance algorithm? Outline the advantages and Limitations.	3	K3	7
UNIT – IV					
7.	a).	Summarize different Free space management techniques used in contiguous memory allocation?	4	K2	6
	b).	What is Inverted Paging? How it different from paging?	4	K2	8
OR					
8.	a).	Find out number of page faults for the given page requests using Least recently Used page replacement algorithm when i) Number of frames =3 ii) Number of Frames = 4 Where, page requests: 1,3,4,0,5,3,2,1,0,4,5,2	4	K3	8
	b).	What is paged segmentation? How it is different from paging?	4	K2	6
UNIT – V					
9.	a).	Consider a disk queue contains cylinder requests for I/O are: 98, 183, 41, 122, 14, 124, 65, 67. The head is initially at cylinder number 53. The cylinders are numbered from 0 to 199. Find out head movement (in number of cylinders) incurred while servicing these requests using i) First Come First Serve disk arm scheduling algorithm. ii) Shortest Seek Time First disk arm scheduling algorithm.	4	K3	8
	b).	What is system protection? Explain different protection mechanisms?	4	K2	6
OR					
10.	a).	What is file system implementation? Explain i-node file allocation algorithm?	4	K2	7
	b).	How the Processes and Files are managed in Windows operating system?	4	K2	7

Course Code: B20IT2202					
SAGI RAMAKRISHNAM RAJU ENGINEERING COLLEGE (A)					R20
II B. Tech II Semester Regular Examinations					
JAVA PROGRAMMING					
(For AIDS & IT)					
Time: 3 Hrs.			Max. Marks:70		
Answer any one Question from Each Unit					
All questions carry equal Marks					
			CO	KL	M
UNIT - I					
1.	a).	Illustrate the concept of JVM with a Diagram.	CO1	K3	7M
	b).	Illustrate the differences between C,C++ and Java with a neat diagram	CO1	K3	8M
OR					
2.	a).	Illustrate the structure of a java program	CO1	K3	8M
	b).	Explain java buzz words in detail.	CO1	K3	7M
UNIT - II					
3.	a).	Write a java program to perform constructor overloading with an example	CO2	K2	8M
	b).	Explain class Declaration syntax in java and modifiers	CO2	K2	7M
OR					
4.	a).	Write a java program to perform method overloading.	CO2	K2	8M
	b).	Explain the keyword final how it is used in java	CO2	K2	7M
UNIT - III					
5.	a).	Explain polymorphism and its types. Construct a java program which Illustrates the functionality of method overloading and method overriding.	CO3	K3	8M
	b).	Write how Multiple Inheritance is possible in java with an example program.	CO3	K3	7M
OR					
6.	a).	Demonstrate an array? Write a java program to read an array of n elements and print them.	CO3	K3	8M
	b).	Write a java program to find second largest number in an array.	CO3	K3	7M
UNIT – IV					
7.	a).	Interpret the concept of packages in java.	CO4	K2	8M
	b).	Construct a java program that shows the functionality of creating a public class in an already existing user defined package.	CO4	K2	7M
OR					
8.	a).	Write a java program to create custom exception.	CO4	K3	8M
	b).	Illustrate 2 different ways of creating a thread in java with code.	CO4	K3	7M
UNIT - V					
9.	a).	Explain any 4 layout mangers in java with example code.	CO5	K2	8M
	b).	Write a java swing code to create a frame and add any five different components onto the frame.	CO5	K3	7M
OR					
10.	a).	Write a java program to explain event handling.	CO5	K2	8M
	b).	Explain JDBC Architecture and procedure to establish JDBC Database Connections.	CO5	K2	7M

			CO	KL	M
UNIT-I					
1	a)	Explain the architecture of 8085 with neat block diagram.	1	2	7
	b)	Explain Different addressing modes in 8085.	1	2	7
OR					
2	a)	Draw and explain signal description of 8085.	1	2	7
	b)	Explain in detail instruction formats and instruction set of 8085	1	2	7
UNIT-II					
3	a)	Write a 8085 program to calculate factorial of a given number	2	3	7
	b)	Explain how instruction execution process occurs in 8085 for any one stack related instruction	2	3	7
OR					
4	a)	Draw timing diagrams for CALL and RETURN instructions.	2	3	7
	b)	Explain the complete interrupt structure of 8085	2	3	7
UNIT-III					
5	a)	Explain interfacing of memory unit with 8085.	3	2	7
	b)	Explain different I/O addressing modes	3	2	7
OR					
6	a)	Draw and explain classification of memory units with a neat sketch.	3	2	7
	b)	Differences between memory mapped I/O and peripheral I/O	3	2	7
UNIT-IV					
7	a)	Explain modes of operation for 8255.	3	2	7
	b)	Explain the block diagram of 8255 and control word format	3	2	7
OR					
8	a)	What is the need of 8254 interface? What are different operation modes of 8254.	3	2	7
	b)	Explain about 8259 interrupt controller operation.	3	2	7
UNIT-V					
9	a)	Draw and explain the format of flag register in 8086.	4	2	7
	b)	Explain the operation modes of 8086 in detail.	4	2	7
OR					
10	a)	Write the importance of BIU & EU functional blocks of 8086.	4	3	7
	b)	Give Addressing modes of 8086 in detail with relevant examples. Calculate physical address for the addressing modes	4	3	7